

# Response to the reviewers for the article entitled *Algorithmic aspects of data-driven computational damage mechanics and introduction of Lipschitz regularization*

September 16, 2026

We thank the reviewers for their careful reading of the manuscript and for the constructive comments. The feedback has led to significant improvements in both the depth and clarity of the work through the addition of new sections and appendices. We have addressed all the points raised, and provide a point-by-point response below. For ease of reference, Reviewer comments are reproduced in italics, followed by our responses. All changes in the revised manuscript are highlighted: additions are shown in blue and deletions in red.

## Reviewer 1

In this paper, the authors adopt the framework of data-driven damage mechanics in which the solution to a boundary value problem is the minimization of a distance between strain and stress pairs with respect to a database subject to the stress-strain pairs satisfying equilibrium. Specifically for this paper, the authors consider the case of damage mechanics in which the stress-strain pairs in the database are history dependent and can exhibit softening. Authors show that the distance function must account for the softening in order to converge. Additionally, the paper tackles the problem of localization. Damage mechanics problems are known to localize to a single element. The regularization introduced is to restrict the norm of the strain to a certain interval. This Lipschitz regularization operates on the strain field because of the nature of the data-driven framework (which must work on the stress-strain field rather than on the elements of the material database), and introduces a length scale that prevents localization of the damage field. The paper is meritorious but some of the details are cumbersome to read and understand and the figure layout is also cumbersome.

1. The role of stress-strain history is not well defined. For example in Eqs. 10, 11 the constitutive model assumes a simple function between strain and stress, no history dependence.
2. 1D example Fig 1 also shows no history dependence. Shouldn't unloading fall back on a line with reduced slope like Fig 10? It seems that this is what is meant by 'elements are forced to stay in the elastic or softening branches' thereby removing any history dependence in this example.
3. Fig 2, Fig 3, Fig 4: There is no need to introduce a figure with just a legend, there is also no need to have figure which is just the zoomed in of a panel in Fig 3. These three figures can easily be consolidated in one.
4. Similar comment for figs 5 and 6.
5. Fig 9 has barely any information.
6. Overall I had a very hard time following the argument surrounding the convergence plots in Figs 3-6 and corresponding text in the article. It seems that a more direct visualization of Eq. 15 and 16 would work better. Also, normalizing with respect to the true value would help seeing when is the algorithm pushing the softening branch toward lower strains (below the max stress strain). How are the branches

forced to stay within their respective branches? That's not clear at all from Eq 12. To be frank, I had a really hard time with this example and corresponding plots. To me, this example is almost better as supplemental material, as it distracts from the more valuable core of the paper stemming from the 1D problem with body force and the 2D problems.

7. 1D Problem with body force. This problem actually introduces the history dependence, as evidenced from Fig 10. However, how is the history dependence handled still not clear. As mentioned before in comment [1], this is a major problem with the current exposition of the method.
8. The rest of the paper is actually much easier to follow, in particular it is clear how the Lipschitz regularization is introduced as a constraint on the norm of the strain gradient. The arc-length is not discussed in detail in the main text but it is well described in Appendix E, where Eq. 61 expresses the constraint in terms of the strain mechanical field. Thus, these extra constraints and regularizations are in terms of the mechanical strain field. The question about how to handle the history dependence on the material stress-strain pairs still bothers me because it is not well covered in the main text.

## Reviewer 2

The paper presents a discussion of the data driven computational approach in the presence of damage localization. To simplify the analysis, instead of material databases, the Authors employ the simple elastic-damage constitutive model. This assumption weakens a bit the paper and the extension of the findings to the real DD method is quite questionable. Still there is an interest in the developed analysis. However, in the reviewer opinion, there are some points that deserve further improvements before possible publication. Hereafter my critical remarks/questions.

1. A first major concern is about the structure of the paper which in my opinion should be significantly revised and improved. In the main text, reference is made to equations coming several pages after. Similarly, the appendices are not sequentially arranged according to their use in the text. Figure 10 is referenced at the beginning of the paper, etc. . . Algorithm 1 suddenly appears at page 20 without any explanation. At page 24 the values of  $Y_c$  and  $k$  are given, but the model was not introduced at this point of the paper. These are some examples of the inconsistency of the structure of the manuscript.
2. Another crucial point concerns the first 1D example (page 8 and 9). In the text it is explained that the procedure converges at slow rate, however there is no common point between mechanical and material states (see figure 3a). What does this mean? It seems a proof of failure of the procedure: please explain.
3. Figures 3, 4, 5, 6 are unclear. If they represent the global response of the whole bar, one would expect just one blue curve of material states and one red of mechanical states. If they refer to the different elements (purple or teal inside the bar), this must be visible. Also, the initial and final points should be marked.
4. The 2D example selected exhibits an extremely brittle behaviour which prevents to follow the evolution of damage with some detail (see figure 12 and 13). Maybe another geometry would help to appreciate the mesh dependence and the following regularization. The sentence at page 16 (line 445) is not justified by the comparison between fig 12d and 12e.
5. Section 4 - Why is the mesh adopted for the strain called "dual" with respect to the one used for displacements? The same order of interpolation is used for displacement and strain, does this ensure the fulfilment of the Babuska-Brezzi conditions? What are the consequences of these interpolations?
6. The removal of the constraint in order to reach failure (zero force), see fig 15, is artificial and put in evidence a non-physical behavior of the model. Also, in the L shape example failure cannot be reached. What is the reason of this?
7. The length  $l_c$  is fixed in an arbitrary way: could you please discuss the issue?

8. The claim at page 28, line 713 that oscillation are reduced is not justified by the results.

Minor remarks:

1. In figure 2 one cannot distinguish the two arrows
2. At page 7, line 214, Blue should be purple
3. At page 22, line 577 the elements cannot be called constant strain triangles since the strain is linearly interpolated
4. Pag 23, line 607, the name of appendix lacks
5. Ref 31 is incomplete